

AI Traffic Signal System

Startup & Daily Operations Guide

Overview

This guide provides step-by-step instructions for setting up the AI-driven Traffic Signal System for the first time, as well as the standard procedure for daily operation.

1 First Time Setup (For New Installations)

Step 1: Hardware Setup:

- Flash the `firmware/traffic_controller/traffic_controller.ino` file to your main ESP32 traffic controller using the Arduino IDE.
- Ensure all three ESP32-CAM modules are powered on and connected to your local Wi-Fi.

Step 2: Software Installation:

- Ensure Python is installed on your computer.
- Open your terminal and install the required dependencies by running:

```
pip install flask requests opencv-python ultralytics
```

Step 3: AI Model Configuration:

- Ensure your custom-trained YOLOv8 model file (`best.pt`) is placed safely inside the `models/` folder.

2 Daily Operations (Routine Use)

1. Power On Hardware:

Turn on the power supply for the main ESP32 traffic pole and the three ESP32-CAMs. Wait roughly 30 seconds for them to connect to your Wi-Fi network.

2. Start the AI Server:

Open your terminal, navigate to the project folder, and run:

```
python app.py
```

Wait until you see `Custom YOLOv8 Model loaded successfully` in the console.

3. Open the Dashboard:

Open your web browser and go to: `http://localhost:5000`.

4. Connect the Systems:

- **Main Controller:** Enter the IP Address of your main ESP32 into the top right input box and click **Connect**.
- **Cameras:** In the cards for Lane 1, Lane 2, and Lane 3, enter the IP addresses for each respective ESP32-CAM and click **Set**.

5. System is Live!

The live video feeds will now appear. The system will automatically begin reading the traffic density (One Car, Two Cars, Three Cars) and dynamically adjusting the green lights based on the traffic load.

3 Emergency Controls

If a crash occurs or maintenance is required, you have 3 override options:

- **Automatic AI:** The AI will detect the "Accident" class and instantly trigger an **All-Red Emergency Mode**.
- **Dashboard Override:** Click the red "Stop All (All Red)" button on the web interface.
- **Physical Override:** Press the physical hardware button on the ESP32 pole to cut power to the lights instantly.

Note: To resume normal traffic after an automatic or dashboard emergency, you must manually click **Clear Emergency** on the dashboard.